

Forensic Analysis of a Suspected Data Leak Using the M57-Jean Disk Image

Khaled Baker

Khalid.baker4961@gmail.com

LinkedIn: [Khalid Baker](#)

GitHub: [KhaledBaker](#)

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Table of Contents

1.	Introduction.....	3
1.1	What is Digital Forensics	3
1.2	Importance of Disk Image Analysis.....	3
1.3	Case Background.....	4
2.	Task 1: Load Forensic Image	4
3.	Task 2: Verify Image Integrity.....	5
4.	Task 3: Explore File System	5
5.	Task 4: User Activity Analysis	6
6.	Task 5: Registry Analysis	7
7.	Task 6: Browser Forensics	8
7.1	Browser History Findings	8
7.2	Cookies Findings	9
7.3	Temporary Internet Files Findings	9
8.	Task 7: File and Metadata Analysis	10
9.	Task 8: Deleted Files Investigation	11
10.	Task 9: Timeline Reconstruction	12
11.	Task 10: Investigation Questions.....	13
11.1	Was the confidential file accessed or modified?	13
11.2	Is there evidence of file transfer exfiltration?	13
11.3	Was any external device used?	13
11.4	What was the last major activity before shutdown?.....	14
11.5	Do you believe Jean's claim? Justify with evidence.....	14
12.	Interpretation of Findings	14
13.	Conclusion	15

1. Introduction

1.1 What is Digital Forensics

Digital forensics is the investigation of digital devices to determine what happened on that device. Digital forensic investigations can examine devices for information like documents, browser history, deleted files, logs, and the traces of external devices that may have been used on that device.

Because most incidents involve computers and digital devices, digital forensics investigations are important in determining the nature of the incident. For example, if there was a data leak from a company, digital forensics investigations could help to determine if the individual that leaked the data was simply accidentally released the data, released the data intentionally, or if some outside third party attacked the company's systems and caused the data to be leaked.

1.2 Importance of Disk Image Analysis

Disk image analysis is one of the most important aspects of digital forensics investigations. A forensic disk image is an exact copy of the devices being imaged, whether they contain active files, deleted files, system files, hidden files or data, and metadata. Investigators often utilize disk images rather than directly investigating the original computer from which the image was taken to ensure that the original evidence is protected from any alteration or damage.

By analyzing the disk image created from the device that is being investigated, the digital investigator can examine the contents of the device while preserving the integrity of the evidence. As digital evidence must be trustworthy, authentic, and in its original state, any alteration of that evidence can lead to the findings from the investigation being less trustworthy. Digital forensic tools, such as FTK Imager, are often used to open and examine the disk image.

Furthermore, disk image analysis allows the investigator to recover any deleted files from the device, examine the timestamps of those files, investigate browser activity, determine what software is installed on the device, and examine system artifacts, such as registry and prefetch files. These system artifacts can assist in forming the timeline of the events that occurred on the device and lead to the conclusion of the investigation.

1.3 Case Background

This project investigates the M57-Jean forensic dataset. The suspected data leak involved a confidential spreadsheet from a company that was leaked online. The file was only believed to be present on Jean's laptop. Furthermore, Jean believes that she was hacked and that the data was leaked as a result of an external attacker. Thus, the investigation will determine if the data leak was the result of an accident, an intention to leak the data, or an external attack.

The investigation will focus upon the forensic disk image of Jean's computer. Within this investigation, the integrity of the computer will be examined for the presence of the data file, the user activity on the computer will be analyzed, and the digital artifacts will be inspected for any evidence of the intentional or accidental data leak. Based upon these findings, a conclusion will be made regarding whether or not Jean's story is believable or if she was at fault for the data leak.

2. Task 1: Load Forensic Image

Firstly, we prepare the forensic image and add it to FTK imager. This is a simple step, all we needed to do is click on File → Add evidence item and select the downloaded file nps-2008-jean.E01. As you can see below the evidence tree shows the selected file indicating that the step was done successfully.



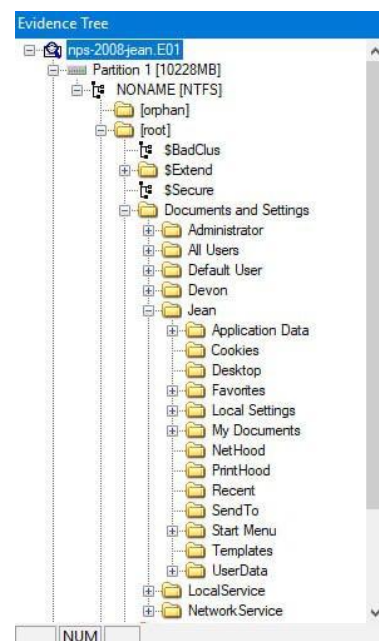
3. Task 2: Verify Image Integrity

For Task 2, the integrity of the image was tested using FTK Imager. The MD5 and SHA1 hash values for the image file nps-2008-jean.E01 were calculated and compared to the verification hash values. The match in the hash values indicates that the image has not been modified, corrupted, or tampered with since it was created. Additionally, FTK Imager also reported that the image contained no bad blocks, indicating that the image is readable. Finally, ensuring the data integrity of the image is important in preserving the authenticity of the digital evidence for the investigation to be valid.

Drive/Image Verify Results	
Name	nps-2008-jean.E01
Sector count	20971520
MD5 Hash	
Computed hash	78a52b5bac78f4e711607707ac0e3f93
Stored verification hash	78a52b5bac78f4e711607707ac0e3f93
Verify result	Match
SHA1 Hash	
Computed hash	ba7dc57e08bb6e3393aee15c713ae04feadc181
Bad Blocks List	
Bad block(s) in image	No bad blocks found in image

4. Task 3: Explore File System

In this task we explored the file system that was created by FTK Imager. The evidence tree reveals the name of the forensic image of the hard drive (nps-2008-jean.E01), the NTFS partition of the drive, and the main directory of the Windows operating system. Navigating into the drive to the folder “Documents and Settings” and selecting the user profile of Jean reveals several folders for Jean, such as Desktop, My Documents, Recent, Cookies, and Local Settings. These folders can reveal information regarding the computer and user activities of Jean. This image is just to show the general outline of the image file and how Jean’s drive looked like. We can also infer that the machine was a windows based machine since it uses the NTFS file system which is used by windows machines. We will dig deeper into this tree to find any evidence of an attack or any possible malicious files.

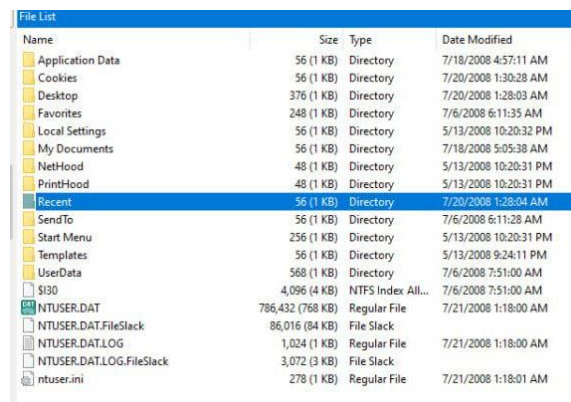


5. Task 4: User Activity Analysis

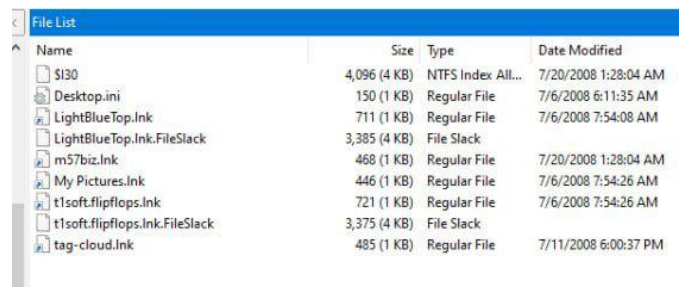
In searching through the users of Jean, we found the spreadsheet file named m57biz.xls within her user folders. This file is relevant to the case scenario of the discovery of a confidential spreadsheet that had been leaked. The location of this file within Jean's user profile indicates that the confidential spreadsheet was stored on Jean's laptop, and that it may have been accessed or utilized by the account. This file can be further analyzed to determine if it was related to the leak of the confidential spreadsheet. This however does not fully indicate that Jean was the leaker it could also mean that an attacker was able to access the system and leak this file. However this does confirm that the file is on Jean's laptop and she cannot claim otherwise. This is a reason to investigate further and find out any relevant information and evidence for this case.



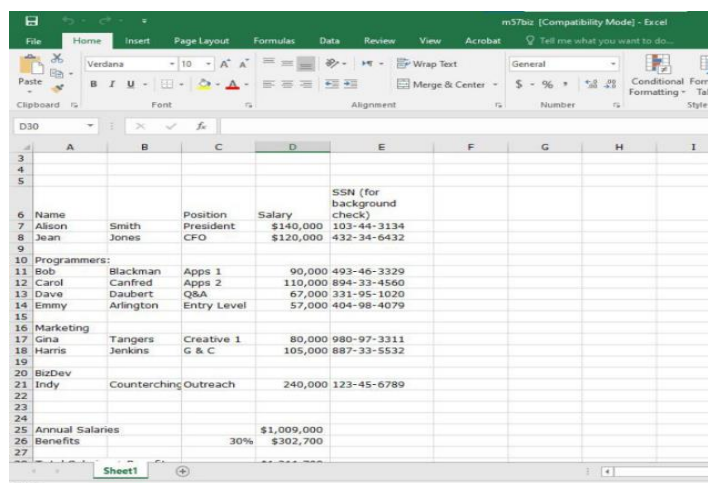
Name	Size	Type	Date Modified
AIM Tunes.url	110 (1 KB)	Regular File	7/18/2008 4:57:11 AM
m57biz.xls	291,840 (285 KB)	Regular File	7/20/2008 1:28:04 AM
m57biz.xls.FileSlack	3,072 (3 KB)	File Slack	



Name	Size	Type	Date Modified
Application Data	56 (1 KB)	Directory	7/18/2008 4:57:11 AM
Cookies	56 (1 KB)	Directory	7/20/2008 1:30:28 AM
Desktop	376 (1 KB)	Directory	7/20/2008 1:28:03 AM
Favorites	248 (1 KB)	Directory	7/6/2008 6:11:35 AM
Local Settings	56 (1 KB)	Directory	5/13/2008 10:20:32 PM
My Documents	56 (1 KB)	Directory	7/18/2008 5:05:38 AM
NetHood	48 (1 KB)	Directory	5/13/2008 10:20:31 PM
PrintHood	48 (1 KB)	Directory	5/13/2008 10:20:31 PM
Recent	56 (1 KB)	Directory	7/20/2008 1:28:04 AM
SendTo	56 (1 KB)	Directory	7/6/2008 6:11:28 AM
Start Menu	256 (1 KB)	Directory	5/13/2008 10:20:31 PM
Templates	56 (1 KB)	Directory	5/13/2008 9:24:11 PM
UserData	568 (1 KB)	Directory	7/6/2008 7:51:00 AM
\$I30	4,096 (4 KB)	NTFS Index All...	7/6/2008 7:51:00 AM
NTUSER.DAT	786,432 (768 KB)	Regular File	7/21/2008 1:18:00 AM
NTUSER.DAT.FileSlack	86,016 (84 KB)	File Slack	
NTUSER.DAT.LOG	1,024 (1 KB)	Regular File	7/21/2008 1:18:00 AM
NTUSER.DAT.LOG.FileSlack	3,072 (3 KB)	File Slack	
ntuser.ini	278 (1 KB)	Regular File	7/21/2008 1:18:01 AM



Name	Size	Type	Date Modified
\$I30	4,096 (4 KB)	NTFS Index All...	7/20/2008 1:28:04 AM
Desktop.ini	150 (1 KB)	Regular File	7/6/2008 6:11:35 AM
LightBlueTop.Ink	711 (1 KB)	Regular File	7/6/2008 7:54:08 AM
LightBlueTop.Ink.FileSlack	3,385 (4 KB)	File Slack	
m57biz.Ink	468 (1 KB)	Regular File	7/20/2008 1:28:04 AM
My Pictures.Ink	446 (1 KB)	Regular File	7/6/2008 7:54:26 AM
t1soft.flipflops.Ink	721 (1 KB)	Regular File	7/6/2008 7:54:26 AM
t1soft.flipflops.Ink.FileSlack	3,375 (4 KB)	File Slack	
tag-cloud.Ink	485 (1 KB)	Regular File	7/11/2008 6:00:37 PM



Name	Position	Salary	SSN (for background check)
Alison Smith	President	\$140,000	103-44-3134
Jean Jones	CFO	\$120,000	432-34-6432
Programmers:			
Bob Blackman	Apps 1	90,000	493-46-3329
Carol Canfred	Apps 2	110,000	894-33-4560
Dave Daubert	QA	67,000	331-95-1020
Emmy Arlington	Entry Level	57,000	404-98-4079
Marketing:			
Cina Tangers	Creative 1	80,000	980-97-3311
Harris Jenkins	G & C	105,000	887-33-5532
BizDev			
Indy Counterchng Outreach		240,000	123-45-6789
Annual Salaries		\$1,009,000	
Benefits	30%	\$302,700	

6. Task 5: Registry Analysis

File List			
Name	Size	Type	Date Modified
default.sav	94,208 (92 KB)	Regular File	5/13/2008 10:19:58 PM
SAM	262,144 (256 KB)	Regular File	7/21/2008 1:31:32 AM
SAM.FileSlack	229,376 (224 KB)	File Slack	
SAM.LOG	1,024 (1 KB)	Regular File	7/21/2008 1:22:18 AM
SAM.LOG.FileSlack	3,072 (3 KB)	File Slack	
SecEvent.Evt	65,536 (64 KB)	Regular File	5/13/2008 10:20:14 PM
SecEvent.Evt.FileSlack	61,440 (60 KB)	File Slack	
SECURITY	262,144 (256 KB)	Regular File	7/21/2008 1:31:32 AM
SECURITY.FileSlack	221,184 (216 KB)	File Slack	
SECURITY.LOG	1,024 (1 KB)	Regular File	7/21/2008 1:22:38 AM
SECURITY.LOG.FileSlack	3,072 (3 KB)	File Slack	
software	12,320,768 (12,...	Regular File	7/21/2008 1:31:32 AM
software.FileSlack	77,824 (76 KB)	File Slack	
software.LOG	1,024 (1 KB)	Regular File	7/21/2008 1:31:17 AM
software.LOG.FileSlack	3,072 (3 KB)	File Slack	
software.sav	659,456 (644 KB)	Regular File	5/13/2008 10:19:58 PM
SysEvent.Evt	196,608 (192 KB)	Regular File	7/21/2008 1:31:12 AM
SysEvent.Evt.FileSlack	45,056 (44 KB)	File Slack	
system	3,932,160 (3,84...	Regular File	7/21/2008 1:31:32 AM
system.FileSlack	57,344 (56 KB)	File Slack	
system.LOG	1,024 (1 KB)	Regular File	7/21/2008 1:22:38 AM
system.LOG.FileSlack	3,072 (3 KB)	File Slack	
system.sav	872,448 (852 KB)	Regular File	5/13/2008 10:19:58 PM
TempKey.LOG	1,024 (1 KB)	Regular File	5/13/2008 10:19:58 PM
TempKey.LOG.FileSlack	3,072 (3 KB)	File Slack	
userdiff	262,144 (256 KB)	Regular File	5/13/2008 10:19:58 PM
userdiff.FileSlack	118,784 (116 KB)	File Slack	
userdiff.LOG	1,024 (1 KB)	Regular File	5/13/2008 10:19:58 PM
userdiff.LOG.FileSlack	3,072 (3 KB)	File Slack	

Export Files...
Export File Hash List...
Add to Custom Content Image (AD1)

Cursor pos = 0; dus = 1760891; log sec = 14087128; phy sec = 14087191

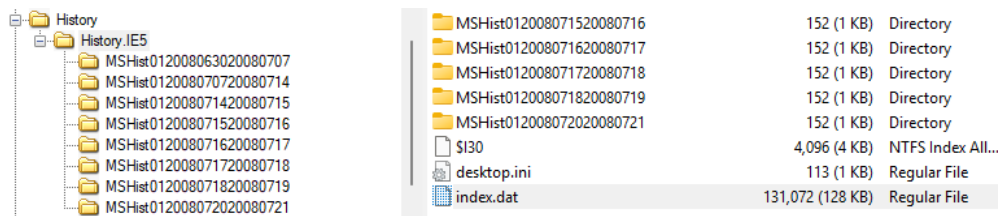
folder l

7. Task 6: Browser Forensics

Browser artifacts were examined within the Internet Explorer folders on Jean's system. The artifacts located within these folders include the History, Cookies, and Temporary Internet Files folder. These artifacts can reveal the type of websites that Jean visited, the activities performed on those websites, the cached webpages that were downloaded to Jean's computer, and the communication services used on that system.

7.1 Browser History Findings

The browser history contains typical browsing activity. However, there are also entries for Google, Live Mail, AIM, and WHOIS activity. While not definitive evidence of a security attack, the browser history does help to show Jean's browsing activity and the type of websites related to communication and accounts that were accessed by Jean.



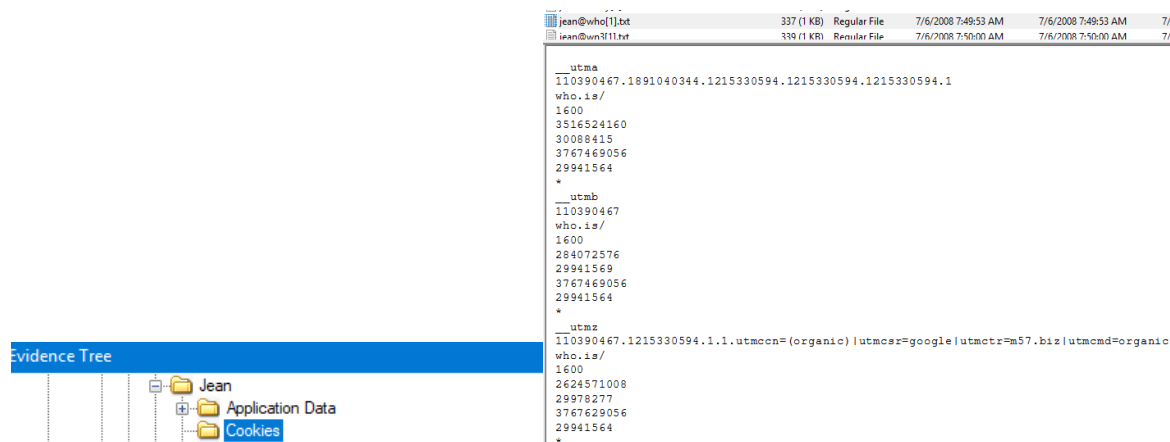
`http://www.who.is/whois_index/m/domain_list.m.0002.phpURL`

`ean@http://www.google.com/search?hl=en&q=m57.biz8m57.biz`

The websites visited include:

- Ebay.com
- Cnn.com
- npr.org
- ms.com
- yahoo.com
- oldnavy.com
- raddison.com

7.2 Cookies Findings



Within the Cookies folder were found a variety of normal cookies that relate to shopping, news, advertisements, and browsing in general. However, some of the cookies that were found were more relevant to the investigation.

The following types of cookies that were located related to Google accounts, Live.com, and AIM/AOL services used by Jean. These cookies indicate that Jean used these types of websites and services. However, they do not imply that Jean leaked the spreadsheet or that an attacker gained access to Jean's computer system.

The finding of the cookie that contained the email address of jean@m57.biz is relevant to the investigation in that it relates to Jean's company account. However, the email address is not suspicious in and of itself, as the m57.biz domain belongs to the company.

7.3 Temporary Internet Files Findings

The Temporary Internet Files folder contained cached webpage files. Such files included normal browsing content as well as files related to the investigation.

Files related to AIM/AOL such as AIM-related pages, AIM cached resources with screen name m57jean, and executable and download artifacts like AIMTunes[1].exe and the AIM installer were located in the Temporary Internet Files folder. Additionally, files related to logging into Microsoft/Live accounts, such as the file WLLogin_JS[1].htm, were also located in this folder.

These files are relevant to the investigation because both AIM/AOL and Live accounts can be used to communicate with others or to transfer files. However, the presence of these files does not indicate that the spreadsheet was uploaded, emailed, or transferred.

8. Task 7: File and Metadata Analysis

For Task 7, we looked into the file metadata and the Windows Prefetch files. There were two main areas of focus for this task: the MAC timestamps for the important spreadsheet file and the Prefetch files for the programs that were executed on the system. The important spreadsheet file was the file named m57biz.xls, which contained the company's confidential spreadsheet and was the file that was likely leaked. For this task, we used the MAC timestamps for the spreadsheet file to determine when it was modified, accessed, and created.

Name	Size	Type	Date Accessed	Date Created	Date Modified
AIM Tunes.url	110 (1 KB)	Regular File	7/20/2008 1:28:02 AM	7/18/2008 4:30:49 AM	7/18/2008 4:30:49 AM
m57biz.xls	291,840 (285 KB)	Regular File	7/20/2008 1:28:03 AM	7/20/2008 1:28:03 AM	7/20/2008 1:28:03 AM
m57biz.xls.FileSlack	3,072 (3 KB)	File Slack			

These timestamps are important to note in that they show the timestamp in which the spreadsheet was published to the system and accessed. The fact that the Created, Modified, and Accessed times are all the same indicates that the spreadsheet was created or copied into this location at the same time that it was accessed. Thus, this timestamp is again helpful in establishing the timeline of events regarding this spreadsheet.

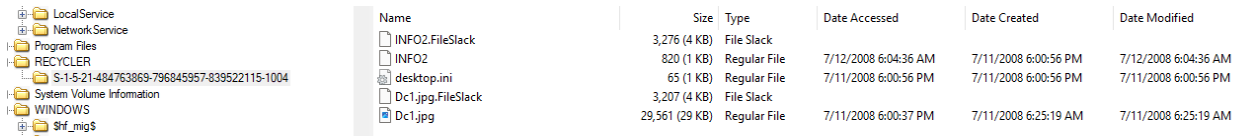
Prefetch Findings

The prefetch folder contained several .pf files showing different programs being executed on the system. The Prefetch files with highest relevance were these:

- CMD.EXE-087B4001.pf — 7/19/2008 8:48:50 AM
- IEXPLORE.EXE-27122324.pf — 7/20/2008 2:59:32 AM
- EXCEL.EXE-1C75F8D6.pf — 7/20/2008 4:27:50 AM
- OUTLOOK.EXE-179DEC04.pf — 7/21/2008 3:44:52 AM
- FIREFOX.EXE-00507E76.pf — 7/21/2008 4:30:44 AM
- AIM6.EXE-34DC5725.pf — 7/21/2008 4:30:46 AM
- AIMINST.EXE-3391E991.pf — 7/18/2008 7:29:14 AM
- AIMTUNES.EXE-07EAB0C7.pf — 7/18/2008 7:30:52 AM

Name	Size	Type	Date Accessed	Date Created	Date Modified
\$I30	53,248 (52 KB)	NTFS Index All...	7/21/2008 1:22:03 AM	6/7/2008 5:05:58 AM	7/20/2008 5:08:28 AM
AGENTSVR.EXE-002E45AB.pf	15,418 (16 KB)	Regular File	7/20/2008 12:00:09 AM	7/6/2008 7:38:56 AM	7/20/2008 12:00:09 AM
AGENTSVR.EXE-002E45AB.pf.FileSlack	966 (1 KB)	File Slack			
AIM6.EXE-34DC5725.pf	93,868 (92 KB)	Regular File	7/21/2008 1:30:46 AM	7/18/2008 4:32:18 AM	7/21/2008 1:30:46 AM
AIMINST.EXE-3391E991.pf	31,896 (32 KB)	Regular File	7/18/2008 4:29:14 AM	7/18/2008 4:26:07 AM	7/18/2008 4:29:14 AM
AIMINST.EXE-3391E991.pf.FileSlack	872 (1 KB)	File Slack			
AIMLANG.EXE-076D401C.pf	21,522 (22 KB)	Regular File	7/18/2008 4:29:17 AM	7/18/2008 4:26:07 AM	7/18/2008 4:29:17 AM
AIMLANG.EXE-076D401C.pf.FileSlack	3,054 (3 KB)	File Slack			
AIMTUNES.EXE-07EAB0C7.pf	15,164 (15 KB)	Regular File	7/18/2008 4:30:52 AM	7/18/2008 4:30:52 AM	7/18/2008 4:30:52 AM
AIMTUNES.EXE-07EAB0C7.pf.FileSlack	1,220 (2 KB)	File Slack			
ALSETUP.EXE-38283348.pf	12,032 (12 KB)	Regular File	7/18/2008 4:28:48 AM	7/18/2008 4:28:48 AM	7/18/2008 4:28:48 AM
AMOS.EXE-219E8D06.pf	26,640 (27 KB)	Regular File	7/18/2008 4:29:24 AM	7/18/2008 4:26:07 AM	7/18/2008 4:29:24 AM
AMOS.EXE-219E8D06.pf.FileSlack	2,032 (2 KB)	File Slack			
AOLDMGR.EXE-2D3B080E.pf	20,462 (20 KB)	Regular File	7/18/2008 4:31:02 AM	7/18/2008 4:30:39 AM	7/18/2008 4:31:02 AM
AOLSOFTWARE.EXE-11870E32.pf	59,904 (59 KB)	Regular File	7/21/2008 1:30:47 AM	7/18/2008 4:32:21 AM	7/21/2008 1:30:47 AM
AOLSOFTWARE.EXE-11870E32.pf.FileSlack	1,536 (2 KB)	File Slack			
AOLTSERVER.EXE-1384CCF8.pf	22,122 (22 KB)	Regular File	7/19/2008 11:59:35 PM	7/18/2008 4:29:31 AM	7/19/2008 11:59:35 PM
AOLTSERVER.EXE-1384CCF8.pf.FileSlack	3,054 (3 KB)	File Slack			

9. Task 8: Deleted Files Investigation



Name	Size	Type	Date Accessed	Date Created	Date Modified
INFO2.FileSlack	3,276 (4 KB)	File Slack			
INFO2	820 (1 KB)	Regular File	7/12/2008 6:04:36 AM	7/11/2008 6:00:56 PM	7/12/2008 6:04:36 AM
desktop.ini	65 (1 KB)	Regular File	7/11/2008 6:00:56 PM	7/11/2008 6:00:56 PM	7/11/2008 6:00:56 PM
Dc1.jpg.FileSlack	3,207 (4 KB)	File Slack			
Dc1.jpg	29,561 (29 KB)	Regular File	7/11/2008 6:00:37 PM	7/11/2008 6:25:19 AM	7/11/2008 6:25:19 AM

For Task 8, the investigation into the files that were deleted on Jean's computer involved accessing the Recycle Bin on the Windows computer. The files in the Recycle Bin on Jean's computer were accessed through the RECYCLER folder.

Files that were deleted by the user on the computer will remain within this folder on the computer. Thus, by examining the contents of this folder, it is possible to discover if any of the deleted files were related to the spreadsheet or the data leak.

Findings

The files discovered within the RECYCLER folder on Jean's computer were:

- Dc1.jpg
- Dc1.jpg.FileSlack
- desktop.ini
- INFO2
- INFO2.FileSlack

The main file that was discovered within the Recycle Bin that was deleted by Jean was the file Dc1.jpg. The file contained a screenshot of the del.icio.us (delicious.com) website and was not related to the spreadsheet.

The file INFO2 is utilized by older versions of the Recycle Bin on Windows computers to store information regarding the deleted files on the computer. Thus, the fact that this file existed indicates that there were deleted files on the computer that could be analyzed.

10. Task 9: Timeline Reconstruction

RE: Please send me the information now Jul 20 2008 04:28am
 From: "Jean User" <jean@m57.biz>
 To: "alison@m57.biz" <tuckgorg@gmail.com>

BEST BODY HEADERS

I've attached the information that you have requested to this email message.

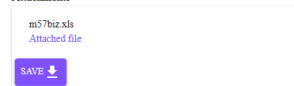
-----Original Message-----
 From: alison@m57.biz [mailto:tuckgorg@gmail.com]
 Sent: Sunday, July 20, 2008 2:23 AM
 To: jean@m57.biz
 Subject: Please send me the information now

Hi, Jean.

I'm sorry to bother you, but I really need that information now --- this VC guy is being very insistent.
 Can you please reply to this email with the information I requested --- the names, salaries, and social security numbers (SSNs) of all our current employees and intended hires?

Thanks.

Alison
 Attachments



This email was found in Jean's outlook email and was discovered by analyzing the outlook.ost file in Jean → Local settings Application Data → Microsoft → Outlook. It appears to be a spear phishing email that was directed to the m57 company, more specifically Jean. This was the cause of the leak and it was done by an attacker that was impersonating "Alison" who appears to be a coworker or supervisor of Jean. Jean was tricked into sending an important and sensitive file containing employee information such as salaries and SSNs. Jean was the leaker however it does not mean she acted maliciously, it just means she lacked the proper training and awareness about phishing attacks.

Date/Time	Evidence Source	Activity
7/11/2008-7/12/2008	Deleted Files/ RECYCLER	Dc1.jpg and INFO2
7/18/2008 7:29:14 AM	Prefetch	AIMINST.EXE-3391E991.pf
7/18/2008 7:30:52 AM	Prefetch	AIMTUNES.EXE-07EAB0C7.pf
7/19/2008 8:48:50 AM	Prefetch	CMD.EXE-087B4001.pf
7/20/2008 1:28:03 AM	File Metadata	m57biz.xls
7/20/2008 2:23 AM	Email Evidence	Original request email sent to Jean.
7/20/2008 2:59:32 AM	Prefetch	IEXPLORE.EXE-27122324.pf
7/20/2008 4:27:50 AM	Prefetch	EXCEL.EXE-1C75F8D6.pf
7/20/2008 4:28 AM	Email Evidence	m57biz.xls
7/21/2008 3:44:52 AM	Prefetch	OUTLOOK.EXE-179DEC04.pf
7/21/2008 4:30:44 AM	Prefetch	FIREFOX.EXE-00507E76.pf
7/21/2008 4:30:46 AM	Prefetch	AIM6.EXE-34DC5725.pf

11. Task 10: Investigation Questions

11.1 Was the confidential file accessed or modified?

The confidential file was accessed on Jean's system. The spreadsheet named m57biz.xls was located on Jean's system and its properties indicate that it was created, modified, and accessed on 7/20/2008 at 1:28:03 AM. This establishes that the file both existed on Jean's computer system and was accessed by Jean at that time. Furthermore, the Prefetch data indicates that Excel was executed on 7/20/2008 at 4:27:50 AM. While this date and time is prior to the indication of the sending of the file via email, the fact that the Excel application was executed on Jean's computer system relates to the investigation of the email.

11.2 Is there evidence of file transfer exfiltration?

There is clear evidence of file transfer or exfiltration through the use of email. The best evidence of such data exfiltration is the finding of an email from Jean that indicated that they were replying to a request for information and attaching the file m57biz.xls to that email. The subject of the email was "RE: Please send me the information now" and the message was sent from Jean User (jean@m57.biz). The email appears to have been sent to an individual that is connected to Alison at M57, but to the email address tuckgorge@gmail.com. The body of the message stated: "I've attached the information that you have requested to this email message." The attached file to this message was named m57biz.xls. Thus, this indicates that the file containing the confidential information of M57 was sent through email from Jean's system.

11.3 Was any external device used?

Given the evidence examined, it is not possible to establish clear proof that an external USB device was used to move the file. While the registry files were located, including the SYSTEM registry hive, there was no evidence of USB device use confirmed within these artifacts. Thus, the evidence supports the claim that the file was emailed rather than moved via external storage.

11.4 What was the last major activity before shutdown?

The last activity to be performed that was related to the data leak was Jean's sending of the email containing the spreadsheet on 7/20/2008 at 4:28 AM. This occurred shortly after the execution of Microsoft Excel at 4:27:50 AM. Afterwards, Jean performed other activities (outlook, firefox, aim), but the most important activity to relate to the data leak was the sending of the spreadsheet via email by Jean.

11.5 Do you believe Jean's claim? Justify with evidence.

Jean's claim that she was hacked is only partly supported by the evidence. The evidence does not indicate that Jean's computer was hacked with some form of malware or other hacking tool. However, the content of the email does indicate that Jean was the victim of a phishing attack or some form of social engineering. The request for the spreadsheet originated from Alison at M57, but the email originated from the external Gmail address of tuckgorge@gmail.com. Furthermore, Jean sent the spreadsheet to Alison along with her reply to the email. Thus, while Jean's computer and email account were involved in the leak of the spreadsheet, the evidence suggests that Jean was deceived into sending the spreadsheet rather than having intentionally leaked the spreadsheet or been hacked into releasing it.

12. Interpretation of Findings

The forensic investigation of the computer image demonstrated that the disk image was successfully loaded, verified, and examined using FTK Imager. The hash of the disk image indicated that the image was reliable and had not been altered in any way. The operating system identified as Windows had an NTFS file system with folders such as Documents and Settings, WINDOWS, Program Files, and RECYCLER. Additionally, Jean's user account and profile was located on the computer that allowed for investigation of her documents, browsing history, registry, deleted files, and applications used.

The most important file that was located during the investigation was the file m57biz.xls, which was the confidential file that was leaked. The file's metadata indicated that it was created, modified, and accessed on the date of 7/20/2008 at 1:28:03 AM. Additionally, the prefetch log files demonstrated that the user utilized Microsoft Excel on 7/20/2008 at 4:27:50 AM. Other programs executed on the computer during this time included Internet Explorer, Outlook, Firefox, and AIM. Furthermore, browsing history on the computer included browsing history regarding Google, Live Mail, AIM/AOL, and general browsing websites. However, there was no evidence of any installed or browsing related to the malicious software on the computer. Additionally, an investigation of the

deleted files on the computer revealed no evidence of anti-forensics investigations by the user. For instance, a file named Dc1.jpg was deleted to the RECYCLER folder and the file could be recovered, but there was no deleted file of the spreadsheet named m57biz.xls.

13. Conclusion

This investigation into the disk image nps-2008-jean.E01 led to the discovery of the method in which the confidential spreadsheet m57biz.xls was leaked. The disk image was loaded into FTK Imager to ensure its integrity. Within the computer that once housed Jean, the spreadsheet was found to exist with a creation date and time of 7/20/2008 at 1:28:03 AM. Additionally, the prefetch data indicated that Excel was executed on the computer shortly before the email transfer of the spreadsheet, indicating its relationship to the transferred data. Furthermore, while the computer did exhibit browsing and communication activity online, no evidence of malware, remote access software, USB data transfer, or hacking was found on the system.

The evidence of the email system indicates that while Jean did respond to the email that requested the spreadsheet, the sender of the email was actually not Alison at M57, but instead the Gmail address of tuckgorge@gmail.com. Additionally, in response to the request for the spreadsheet, Jean included an email attachment of the spreadsheet named m57biz.xls. These findings indicate that Jean's computer and email were linked to the spread of the spreadsheet. However, no evidence suggests that Jean had intentionally leaked the spreadsheet. Instead, the findings suggest that Jean was likely deceived by the email that requested the spreadsheet. Thus, the findings of this investigation indicate that the spreadsheet was spread due to a deceived email response, not as a result of intentional malicious actions.